Hush, mice...

W21 (2021) — English translation

Seeing Leopold standing on the high edge of the ravine from the opposite bank, the mice threw a stone straight at him at a speed of v_1. Leopold noticed this and, at the moment of the throw, jumped with a speed of v_2 in the plane of the stone's flight path perpendicular to the line connecting it with the mice (see figure). When the distance between Leopold and the stone became minimal, their speeds were again perpendicular.

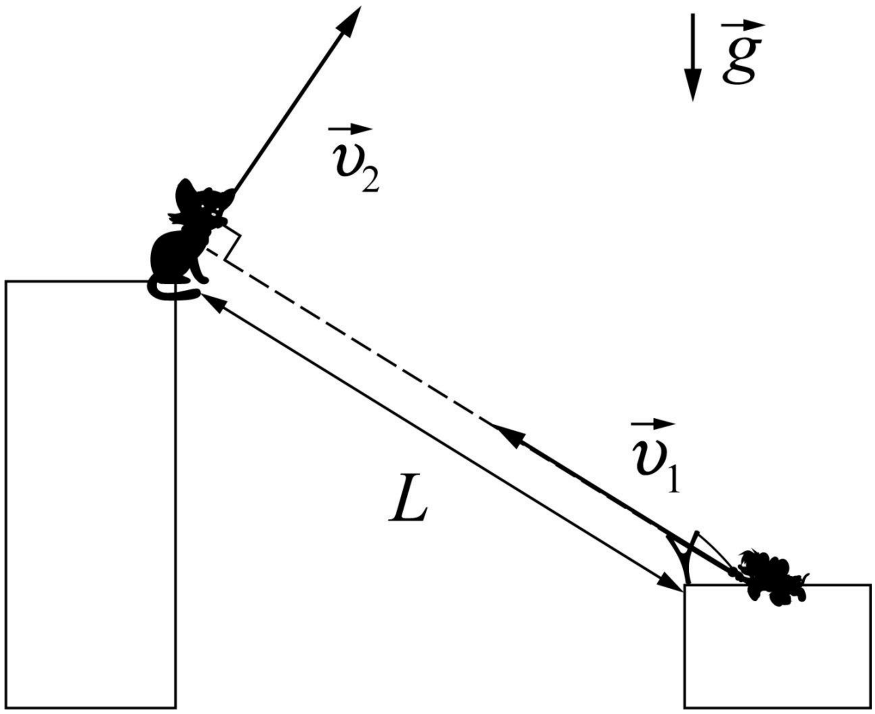

Figure 1: Figure 1.

A1	Find the maximum possible initial distance L_{max} between the mice and Leopold.	**5.00**

When the distance between Leopold and the stone became minimal, Leopold, the mice and the stone were photographed. The initial distance between the mice and Leopold was the maximum possible. In the figure, the point A denotes the position of Leopold, B the position of the mice, C the stone at the time of the photograph. The orientation of the photo is unknown. You can use a graduated ruler.

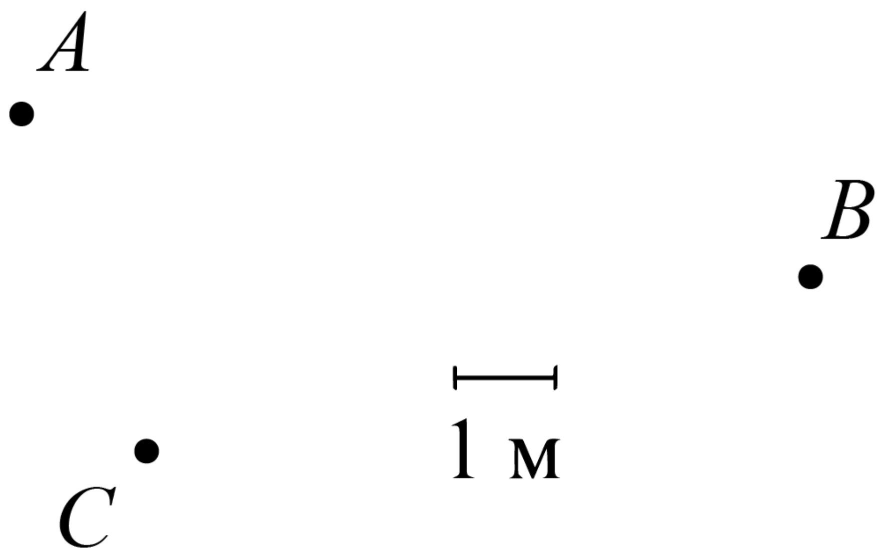

Figure 2: Figure 2.

A2	Find the values of the speeds v_1 and v_2. Gravity acceleration $g = 9,8$ m/s^2.	**5.00**

Source: `https://pho.rs/p/69`